 **Enrollment No :** ____________________

# Faculty of Engineering / Science

# End Sem Examination Dec 2025

## CS3CO31 / BC3CO36 Data Structures (T)

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|---|---|---|---|---|---|
| **Programme** | : | B. Tech. / B. Sc. (Hons.) | **Branch/Specialisation** | : | CSE All / CS |
| **Duration** | : | 3 hours | **Maximum Marks** | : | 60 |

**Note:** *All questions are compulsory. Internal choices, if any, are indicated. Assume suitable data if necessary. Notations and symbols have their usual meaning.*

### Section 1 (Answer all question(s))

**Marks CO BL**

**Q1.** Differentiate persistent and non-persistent data structure. **2 1 1**

| Rubric | Marks |
|---|---|
| Differentiate Persistent and Non-Persistent data structure. | 2 |

**Q2.** Trace the recursive function fun(n) defined as follows and find the output for fun(5).: **4 2 2**

```
int fun(int n){
  if(n <= 1) return 1;
  else return n * fun(n-1);
}
```

| Rubric | Marks |
|---|---|
| Write complete flow of the function call - 3 marks<br>Only Answer- 1 mark | 4 |

**Q3.** **(a)** Write a C program to insert an element into an array at specific position given by the user. **6 5 5**

| Rubric | Marks |
|---|---|
| Write a C program to insert an element into an array at specific position given by the user. | 6 |

**(OR)**

**(b)** Write a C program to delete an element from an array at specific position given by the user.

| Rubric | Marks |
|---|---|
| Write a C program to delete an element from an array at specific position given by the user. | 6 |

### Section 2 (Answer all question(s))

**Marks CO BL**

**Q4.** What are the advantages of linked lists over arrays? **2 1 1**

| Rubric | Marks |
|---|---|
| What are the advantages of linked lists over arrays? | 2 |

**Q5.** Write a C structure definition for a doubly linked list node and create two nodes properly connected in the doubly linked list. The data values to be stored in these nodes may taken by the user or may be given static value in the program itself. **4 4 4**

| Rubric | Marks |
|---|---|
| Structure - 1 mark<br>one node -1 mark<br>another node - 1 mark<br>connection - 1 mark | 4 |

**Q6. (a)** Write a C program to generate a Singly linked list dynamically for n number of nodes. The value of n should be given by the user at run time. Display the values of the nodes of the linked list after creation of the linked list. **6 5 5**

| Rubric | Marks |
|---|---|
| Structure -1 mark<br>Creation of dynamic linked list- 3 mark<br>Display- 2 marks | 6 |

**(OR)**

**(b)** What are applications of circular linked list? How is it different from a linear linked list? Write an algorithm to traverse a circular linked list.

| Rubric | Marks |
|---|---|
| Applications - 1<br>Difference -2<br>Algorithm - 3 | 6 |

## Section 3 (Answer all question(s))

**Marks CO BL**

**Q7.** Differentiate circular queue and priority queue. **2 2 2**

| Rubric | Marks |
|---|---|
| Differentiate circular queue and priority queue. | 2 |

**Q8.** Find postfix and prefix expression of the following expression using a stack: **4 3 3**
(A+B)*((C-D)/E)
Show steps properly.

| Rubric | Marks |
|---|---|
| 2 marks for each | 4 |

**Q9. (a)** Write push and pop functions for a stack in C programming language. **6 5 5**

| Rubric | Marks |
|---|---|
| 3 marks each | 6 |

**(OR)**

**(b)** Write insert and delete functions of queue in C programming language.

| Rubric | Marks |
|---|---|
| 3 marks each | 6 |

## Section 4 (Answer all question(s))

**Marks CO BL**

**Q10.** Explain radix sort. **2 1 1**

| Rubric | Marks |
|---|---|
| Explanation | 2 |

**Q11.** Define hashing with example. Explain collision resolution techniques. **4 2 2**

| Rubric | Marks |
|---|---|
| Definition-1<br>Example -1<br>CRT-2 | 4 |

**Q12. (a)** Explain bubble sort technique for sorting an array. Write each and every steps of sorting for the following list:
15, 61, 8, 20, 5, 101. **6 3 3**

| Rubric | Marks |
|---|---|
| Explanation -2 marks<br>Sorting- 4 marks | 6 |

**(OR)**

**(b)** Write a difference between linear search & binary search. Explain using any example.

| Rubric | Marks |
|---|---|
| Difference -3 marks<br>examples 3 marks | 6 |

## Section 5 (Answer all question(s))

**Marks CO BL**

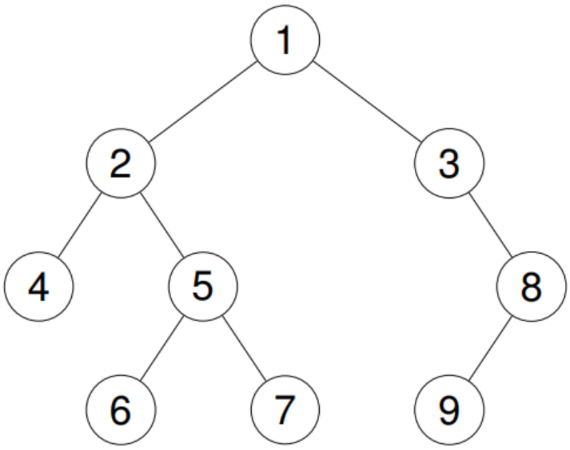
**Q13.** Write any two differences between tree and graph data structures. **2 2 2**

| Rubric | Marks |
|---|---|
| Diff - 2 | 2 |

**Q14.** Insert 50, 30, 70, 20, 40, 60, 80 into a binary search tree and draw the resulting tree. **4 3 3**

| Rubric | Marks |
|---|---|
| Insert **50, 30, 70, 20, 40, 60, 80** into a Binary Search Tree and draw the resulting tree. | 4 |

**Q15. (a)** **6 4 4**




Find inorder, preorder and postorder traversals of the tree given in the figure. Show steps properly.

| Rubric | Marks |
| --- | --- |
| 2 marks each | 6 |

**(OR)**

**(b)** What are different graph traversal techniques? Explain the traversal techniques with an example. Also write one applications of each.

| Rubric | Marks |
| --- | --- |
| BFS- 3 Marks, DFS- 3 Marks | 6 |

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